

Pracownia Systemów i Urządzeń Sieciowych		
03.11.2022 - Automatyzacja tworzenia użytkowników		
	Imię i Nazwisko	klasa/grupa/ rok szk.
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INTERFEJSY

ST - LAN1 - zintegrowany

LAN2 - na porcie PCIe

S - LAN1 - zintegrowany

LAN2 - na porcie PCIe

TREŚĆ ZADANIA

1. Sprawdzanie informacji o obiektach za pomocą **ldifde**.
2. Tworzenie i usuwanie kont użytkowników za pomocą **ldifde**.
3. Tworzenie i usuwanie kont użytkowników za pomocą **Windows PowerShell**.

1. SPRAWDZANIE INFORMACJI O OBIEKTACH ZA POMOCĄ ldifde

INFORMACJE O KOMENDZIE

```
Microsoft Windows [Version 10.0.17763.2028]
(c) 2018 Microsoft Corporation. Wszelkie prawa zastrzeżone.

C:\Users\Administrator>ldifde

LDIF Directory Exchange

General Parameters
=====
-i Turn on Import Mode (The default is Export)
-f filename Input or Output filename
-s servername The server to bind to (Default to DC of computer's domain)
-t FromDN ToDN Replace occurrences of FromDN to ToDN
-r If either FromDN or ToDN ends with #attributeName, the
attribute value will be looked up in rootDSE and used to
replace #attributeName. See example for "Macro expansion
in DN's".
-v Turn on Verbose Mode
-l path Log File Location
-p port Port Number (default = 389)
-u Use Unicode format
-w timeout Terminate execution if the server takes longer than the
specified number of seconds to respond to an operation
(default = no timeout specified)
-h Enable SASL layer signing and encryption
-j Help

Export Specific
=====
-d RootDN The root of the LDAP search (Default to Naming Context)
-f filter LDAP search filter (Default to "(objectclass=*)")
-s SearchScope Search Scope (Base/Onelevel/Subtree)
-i list List of attributes (comma separated) to look for
in an LDAP search
-o list List of attributes (comma separated) to omit from
input.
-g Disable Paged Search.
-m Enable the SAM logic on export.
-n Do not export binary values
-x Include deleted objects (tombstones)
-l Retain only the important replPropertyMetadata

Import
=====
-k The import will go on ignoring 'Constraint Violation'
and 'Object Already Exists' errors
-y The import will use lazy commit for better performance
(enabled by default)
-e The import will not use lazy commit
-q threads The import will use the specified number of threads
(default is 1)
-z Continue importing irrespective of errors.
-x Enable tombstone reanimation support (passes deleted
objects control with ldap modify requests)

Credentials Establishment
=====
Note that if no credentials is specified, LDIFDE will bind as the currently
logged on user, using SSPI.
-a UserDN [Password | *] Simple authentication
-b UserName Domain [Password | *] SSPI bind method

Example: Simple import of current domain
ldifde -i -f INPUT.LDF

Example: Simple export of current domain
```

```
-x Include deleted objects (tombstones)
-l Retain only the important replPropertyMetadata

Import
=====
-k The import will go on ignoring 'Constraint Violation'
and 'Object Already Exists' errors
-y The import will use lazy commit for better performance
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-z Continue importing irrespective of errors.
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objects control with ldap modify requests)

Credentials Establishment
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Note that if no credentials is specified, LDIFDE will bind as the currently
logged on user, using SSPI.
-a UserDN [Password | *] Simple authentication
-b UserName Domain [Password | *] SSPI bind method

Example: Simple import of current domain
ldifde -i -f INPUT.LDF

Example: Simple export of current domain
ldifde -f OUTPUT.LDF

Example: Export of specific domain with credentials
ldifde -m -f OUTPUT.LDF
-b USERNAME DOMAINNAME *
-s SERVERNAME
-d "cn=users,DC=DOMAINNAME,DC=Microsoft,DC=Com"
-r "(objectclass=user)"

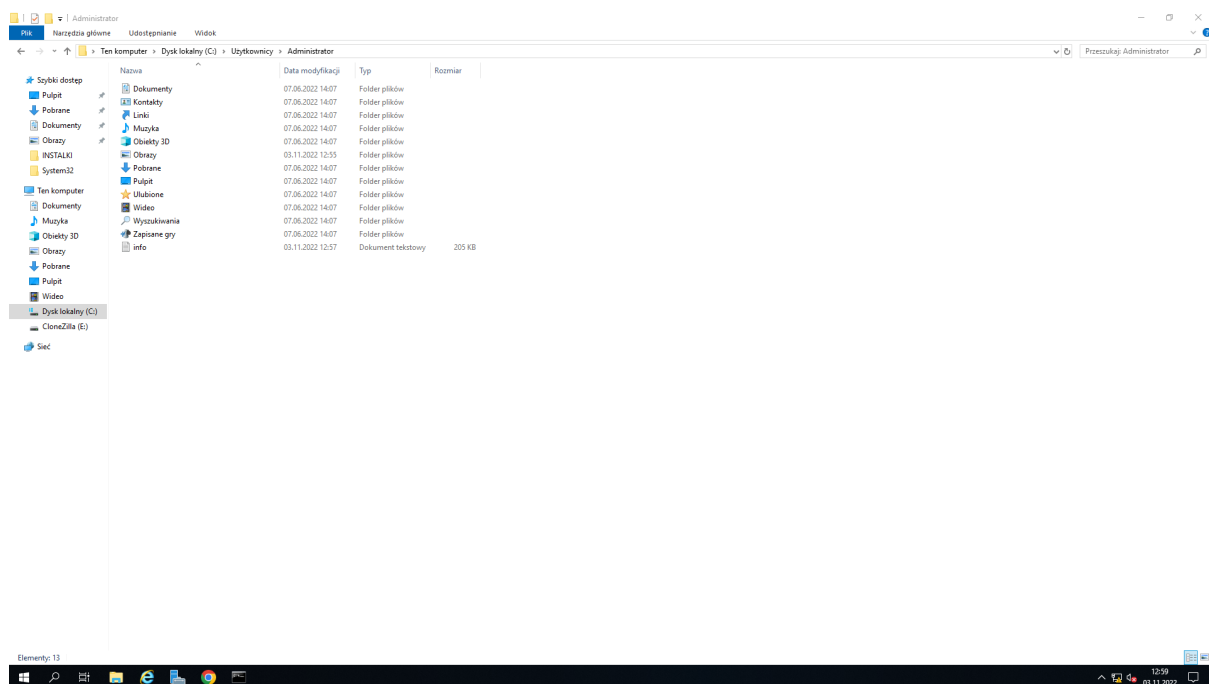
Example: Macro expansion in DN's
ldifde -f export.ldf -c "#configurationNamingContext" "cn=configuration,dc=com"
ldifde -i -f import.ldf -c "cn=configuration,dc=com" "#configurationNamingContext"

No log files were written. In order to generate a log file, please
specify the log file path via the -j option.

C:\Users\Administrator>
```

TRYB IMPORTU

```
C:\Users\Administrator>idfde -f info.txt
Connecting to "WIN-U96UWH34HA0.zst.com"
Logging in as current user using SSP!
Exporting directory to file info.txt
Searching for entries...
Writing out entries.....
.....
241 entries exported
.....
The command has completed successfully
C:\Users\Administrator>
```



Plik tekstowy możemy odnaleźć w katalogu głównym administratora

```
info - Notatnik
Plik Edycja Format Widok Pomoc
dn: DC=zst,DC=com
changetype: add
objectClass: top
objectClass: domain
objectClass: domainDNS
distinguishedName: DC=zst,DC=com
instanceType: 5
whenCreated: 20221103113821.02
whenChanged: 20221103114236.02
subRefs: DC=ForestDnsZones,DC=zst,DC=com
subRefs: DC=DomainDnsZones,DC=zst,DC=com
subRefs: CN=Configuration,DC=zst,DC=com
uSNCreated: 4099
uSNChanged: 12583
name: zst
objectGUID: 1b00eV0AGF+u2C9c1Zrzuw+
creationTime: 133119492835879287
forceLogoff: -9223372036854775808
lockoutDuration: -18000000000
lockoutObservationWindow: -18000000000
lockoutThreshold: 0
maxPwdAge: -36288000000000
minPwdAge: -8640000000000
minPwdLength: 7
modifiedCountAtLastProm: 0
nextRid: 1000
pwdProperties: 1
pwdHistoryLength: 24
objectSID: AQQAAAAAUUAAAAwL3BRVtzj0ok998n
serverState: 1
uASCompat: 1
modifiedCount: 1
auditingPolicy: AAE-
nTLMixedDomain: 0
rIDManagerReference: CN=RID Manager$,CN=System,DC=zst,DC=com
FSRMRoleOwner:
  CN=NTDS Settings,CN=NTDS-U96UPM14HAD,CN=Servers,CN=Default-First-Site-Name,CN=5
  ites,CN=Configuration,DC=zst,DC=com
systemFlags: -1946157056
wellKnownObjects:
  B:32:6227F0AF1FC241008E38B106158B580F:CN=NTDS Quotas,DC=zst,DC=com
wellKnownObjects:
  B:32:F48E92A4C777485E878E9421D530870B:CN=Microsoft, CN=Program Data,DC=zst,DC=c
  om
wellKnownObjects:
  B:32:09460C0BAE1E4AAE48F64AE7DAA1E5A:CN=Program Data,DC=zst,DC=com
wellKnownObjects:
  B:32:22870C67D56E4EFB91E9300FCA3DC1AA:CN=ForeignSecurityPrincipals,DC=zst,DC=c
  om
wellKnownObjects:
  B:32:18E2EA08684F1D2B9AA00C04F79F805:CN=Deleted Objects,DC=zst,DC=com
wellKnownObjects:
  B:32:2FBAC1870ADE11D297C400C04F0805CD:CN=Infrastructure,DC=zst,DC=com
```

Możemy tu znaleźć wszystkie informacje o naszym serwerze

UZYSKANIE INFORMACJI O KONKRETNYM OBIEKCIE

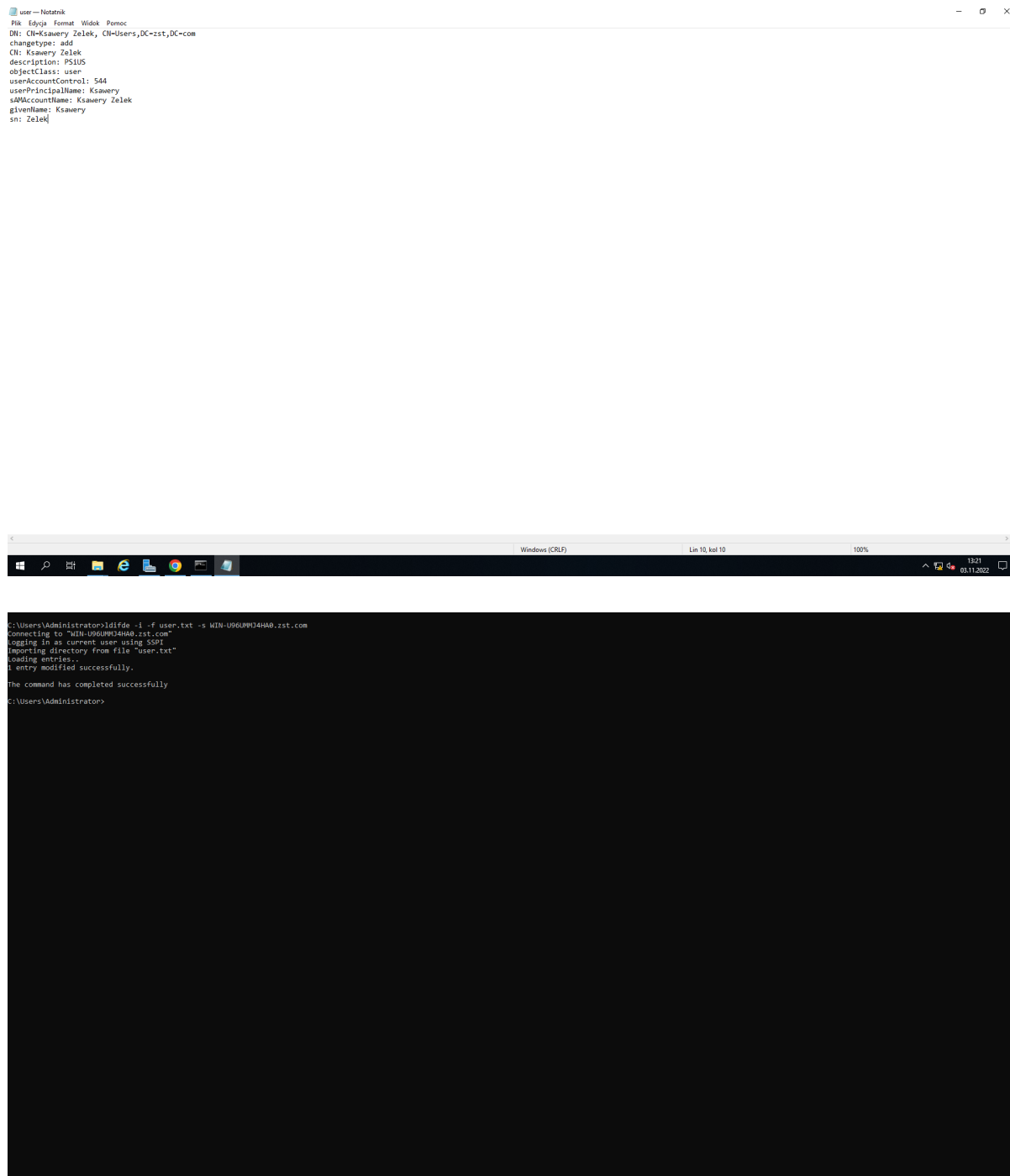
```
C:\Users\Administrator>ldifde -f info2.txt -s WIN-U96UWD4HA0 -d "CN=user, CN=Users, DC=zst, DC=com"
Connecting to "WIN-U96UWD4HA0"
Logging in as current user using SSL
Exporting directory to file info2.txt
Searching for entries...
Writing out entries.
1 entries exported
The command has completed successfully
C:\Users\Administrator>
```

```
info2 - Notatnik
Plk: Edycja Format Widok Pomoc
dn: CN=user, CN=Users, DC=zst, DC=com
changetype: add
objectClass: top
objectClass: person
objectClass: organizationalPerson
objectClass: user
cn: user
givenName: user
distinguishedName: CN=user, CN=Users, DC=zst, DC=com
instancetype: 4
whenCreated: 20221103120352.0Z
whenChanged: 20221103120352.0Z
displayName: user
uSNCreated: 12824
uSNChanged: 12829
name: user
objectGUID:: x149Wk1RkMygIT033ocua==
userAccountControl: 66048
badPwdCount: 0
codePage: 0
countryCode: 0
badPasswordTime: 0
lastLogoff: 0
lastLogon: 0
pwdLastSet: 131119506321644566
primaryGroupID: 513
objectSid:: AQUIAAAAAAAAAAAAAw13BRVtzJ0ak99RnTwQAAA==
accountExpires: 9223372036854775807
logonCount: 0
sAMAccountName: user
sAMAccountType: 805306368
userPrincipalName: user@zst.com
objectCategory: CN=Person, CN=Schema, CN=Configuration, DC=zst, DC=com
dScorePropagationData: 16010101000000.0Z
```

2. TWORZENIE I USUWANIE KONT UŻYTKOWNIKÓW ZA POMOCĄ Idifde

Warto wspomnieć, że używając tej komendy mamy ograniczone możliwości, ale możemy zakładać konta dla większej grupy użytkowników.

UTWORZENIE PLIKU TEKSTOWEGO Z DANYMI UŻYT.



```
user - Notepad++
Plik Edycja Format Widok Pomoc
DN: CN=Ksawery Zelek, CN=Users, DC=zst, DC=com
changetype: add
CN: Ksawery Zelek
description: PSUS
objectClass: user
userAccountControl: 544
userPrincipalName: Ksawery
sAMAccountName: Ksawery Zelek
givenName: Ksawery
sn: Zelek

C:\Users\Administrator>idifde -i -f user.txt -s WIN-UP6UPD4H40.zst.com
Connecting to "WIN-UP6UPD4H40.zst.com"
Loading in as current user using SSO
Importing directory from file "user.txt"
Loading entries...
1 entry modified successfully.
The command has completed successfully
C:\Users\Administrator>
```

TWORZENIE WIELU UŻYTKOWNIKÓW

```
user - Notatnik
Plik Edycja Format Widok Pomoc
description: PS1US
objectClass: user
userAccountControl: 544
userPrincipalName: Ksawery10
sAMAccountName: Ksawery Zelek10
givenName: Ksawery10
sn: Zelek10

DN: CN=Ksawery Zelek1, CN=Users,DC=zst,DC=com
changetype: add
CN: Ksawery Zelek1
description: PS1US
objectClass: user
userAccountControl: 544
userPrincipalName: Ksawery1
sAMAccountName: Ksawery Zelek1
givenName: Ksawery1
sn: Zelek1

DN: CN=Ksawery Zelek2, CN=Users,DC=zst,DC=com
changetype: add
CN: Ksawery Zelek2
description: PS1US
objectClass: user
userAccountControl: 544
userPrincipalName: Ksawery2
sAMAccountName: Ksawery Zelek2
givenName: Ksawery2
sn: Zelek2

DN: CN=Ksawery Zelek3, CN=Users,DC=zst,DC=com
changetype: add
CN: Ksawery Zelek3
description: PS1US
objectClass: user
userAccountControl: 544
userPrincipalName: Ksawery3
sAMAccountName: Ksawery Zelek3
givenName: Ksawery3
sn: Zelek3

DN: CN=Ksawery Zelek4, CN=Users,DC=zst,DC=com
changetype: add
CN: Ksawery Zelek4
description: PS1US
objectClass: user
userAccountControl: 544
userPrincipalName: Ksawery4
sAMAccountName: Ksawery Zelek4
givenName: Ksawery4
sn: Zelek4

DN: CN=Ksawery Zelek5, CN=Users,DC=zst,DC=com
```

```
C:\Users\Administrator>ldifde -i -f user.txt -s WIN-U96UPM34HA0.zst.com
Connecting to "WIN-U96UPM34HA0.zst.com"
Logging in as current user using SSP
Importing directory from file "user.txt"
Loading entries.....
10 entries modified successfully.

The command has completed successfully
C:\Users\Administrator>
```

USUWANIE WIELU UŻYTKOWNIKÓW

```
user - Notatnik
Plik Edycja Format Widok Szukaj
DN: CN=Ksawery Zelek,CN=Users,DC=zst,DC=com
changetype: delete

DN: CN=Ksawery Zelek1,CN=Users,DC=zst,DC=com
changetype: delete

DN: CN=Ksawery Zelek2,CN=Users,DC=zst,DC=com
changetype: delete

DN: CN=Ksawery Zelek3,CN=Users,DC=zst,DC=com
changetype: delete

DN: CN=Ksawery Zelek4,CN=Users,DC=zst,DC=com
changetype: delete

DN: CN=Ksawery Zelek5,CN=Users,DC=zst,DC=com
changetype: delete

DN: CN=Ksawery Zelek6,CN=Users,DC=zst,DC=com
changetype: delete

DN: CN=Ksawery Zelek7,CN=Users,DC=zst,DC=com
changetype: delete

DN: CN=Ksawery Zelek8,CN=Users,DC=zst,DC=com
changetype: delete

DN: CN=Ksawery Zelek9,CN=Users,DC=zst,DC=com
changetype: delete

DN: CN=Ksawery Zelek10,CN=Users,DC=zst,DC=com
changetype: delete

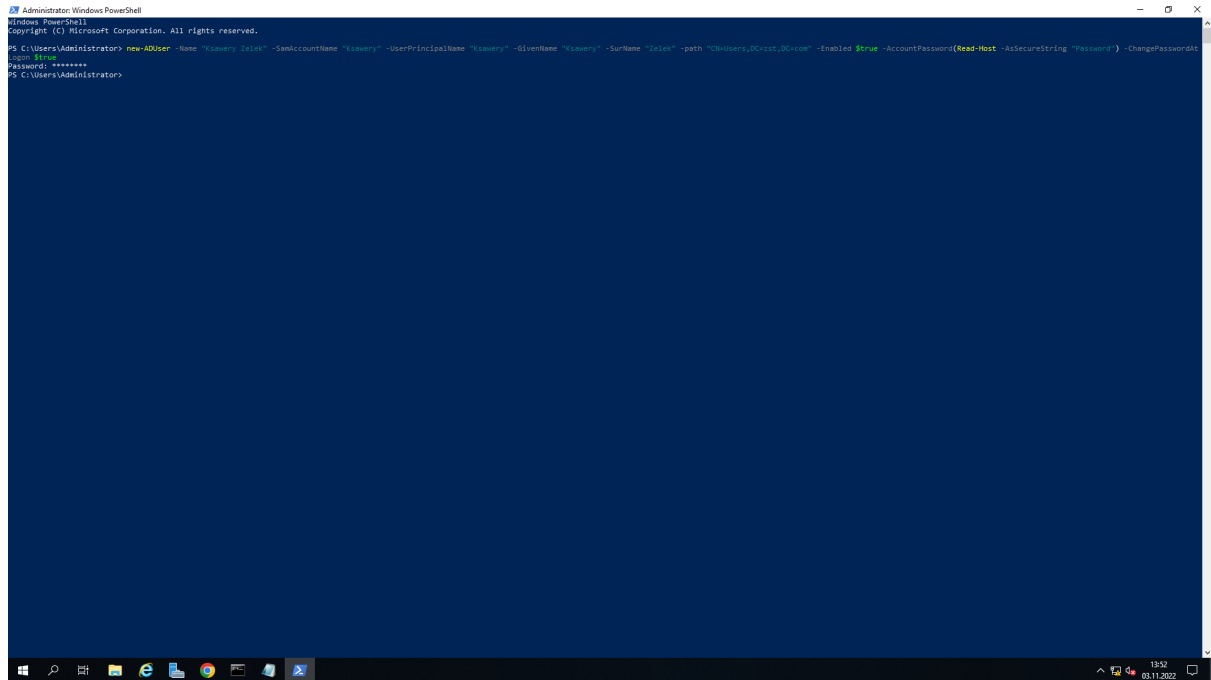
C:\Users\Administrator>ldifde -i -f user.txt -s WIN-U96UPM34HA0.zst.com
Connecting to "WIN-U96UPM34HA0.zst.com"
Logging in as current user using SSP
Importing directory from file "user.txt"
Loading entries.....
11 entries modified successfully.

The command has completed successfully
C:\Users\Administrator>
```


3. TWORZENIE I USUWANIE KONT UŻYTKOWNIKÓW ZA POMOCĄ Windows PowerShell

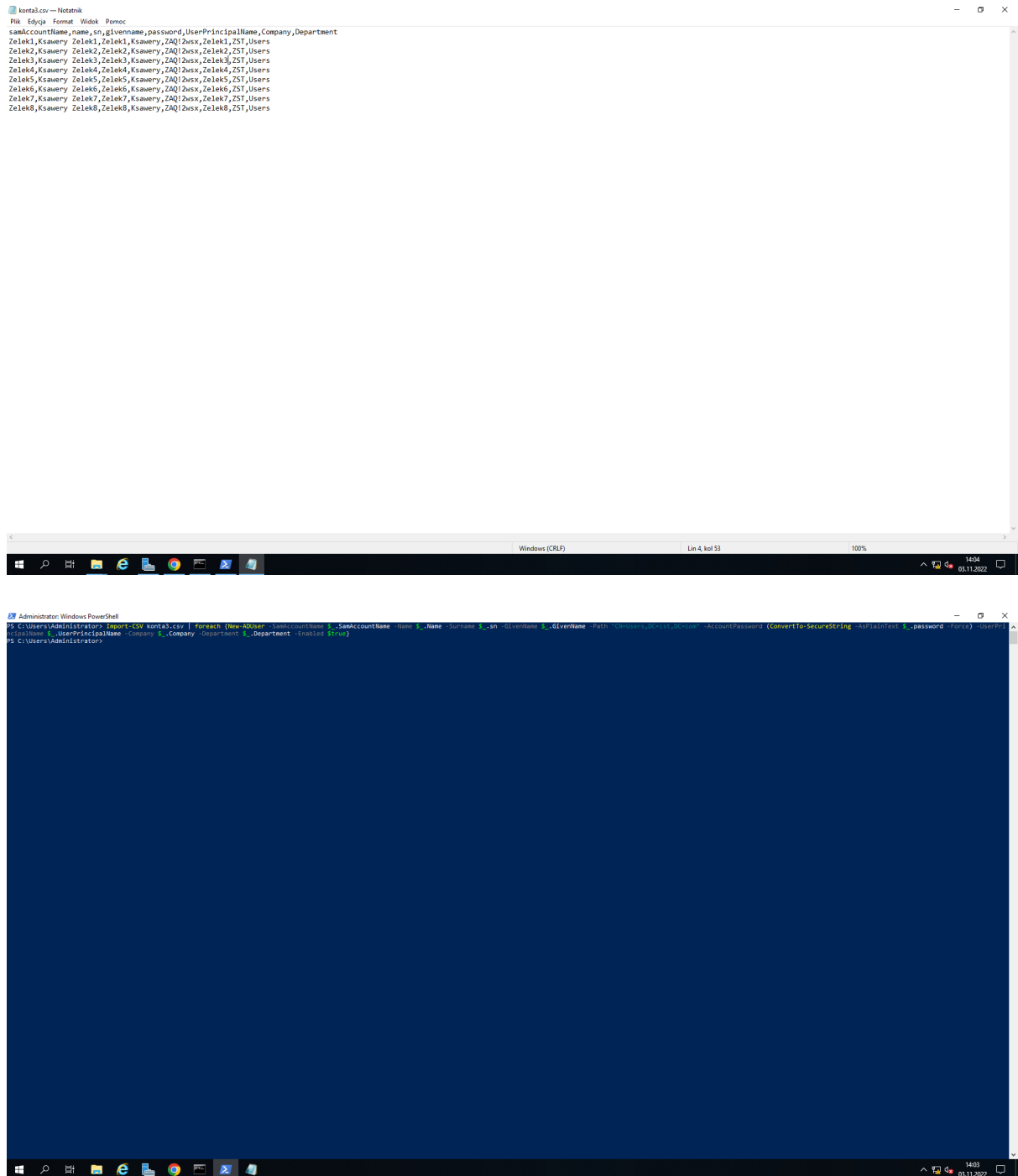
To narzędzie ma większe możliwości niż polecenie **ldifde**.

TWORZENIE UŻYTKOWNIKA W POWERSHELL

A screenshot of a Windows PowerShell terminal window titled "Administrator: Windows PowerShell". The window shows the command to create a new Active Directory user. The command is: `PS C:\Users\Administrator> new-ADUser -Name "Ksawery Zelek" -SamAccountName "Ksawery" -UserPrincipalName "Ksawery" -GivenName "Ksawery" -SurName "Zelek" -path "CN=Users,DC=zst,DC=com" -Enabled $true -AccountPassword(Read-Host -AsSecureString "Password") -ChangePasswordAtLogon $true`. The terminal background is dark blue. The Windows taskbar is visible at the bottom with the date 09.11.2022 and time 13:52.

```
PS C:\Users\Administrator> new-ADUser -Name "Ksawery Zelek"
-SamAccountName "Ksawery" -UserPrincipalName "Ksawery" -GivenName
"Ksawery" -SurName "Zelek" -path "CN=Users,DC=zst,DC=com" -Enabled
$true -AccountPassword(Read-Host -AsSecureString "Password")
-ChangePasswordAtLogon $true
```

DODAWANIE WIĘKSZEJ ILOŚCI UŻYTKOWNIKÓW



The image shows a Windows desktop environment. The top window is Notepad++ editing a file named 'konta3.csv'. The file contains a CSV header and several rows of user data. The bottom window is an Administrator Windows PowerShell terminal. It shows the command to import the CSV file into Active Directory using the 'Import-CSV' and 'foreach' cmdlets, along with various parameters for user creation.

```
konta3.csv -- Notatnik
plik Edycja Format Widok Pomoc
samaccountname,name,sn,givenname,password,userPrincipalName,company,department
Zelek1,Ksawery Zelek1,Zelek1,Ksawery,ZAQ12wsx,Zelek1,ZST,Users
Zelek2,Ksawery Zelek2,Zelek2,Ksawery,ZAQ12wsx,Zelek2,ZST,Users
Zelek3,Ksawery Zelek3,Zelek3,Ksawery,ZAQ12wsx,Zelek3,ZST,Users
Zelek4,Ksawery Zelek4,Zelek4,Ksawery,ZAQ12wsx,Zelek4,ZST,Users
Zelek5,Ksawery Zelek5,Zelek5,Ksawery,ZAQ12wsx,Zelek5,ZST,Users
Zelek6,Ksawery Zelek6,Zelek6,Ksawery,ZAQ12wsx,Zelek6,ZST,Users
Zelek7,Ksawery Zelek7,Zelek7,Ksawery,ZAQ12wsx,Zelek7,ZST,Users
Zelek8,Ksawery Zelek8,Zelek8,Ksawery,ZAQ12wsx,Zelek8,ZST,Users

Administrator: Windows PowerShell
PS C:\Users\Administrator> Import-CSV kontas3.csv | foreach {New-ADUser -SamAccountName $_.SamAccountName -Name $_.Name -Surname $_.sn -GivenName $_.GivenName -Path "CN=Users,DC=zst,DC=com" -AccountPassword (ConvertTo-SecureString -AsPlainText $_.password -Force) -UserPrincipalName $_.UserPrincipalName -Company $_.Company -Department $_.Department -Enabled $true}
```

```
PS C:\Users\Administrator> Import-CSV kontas3.csv | foreach {New-ADUser
-SamAccountName $_.SamAccountName -Name $_.Name -Surname $_.sn
-GivenName $_.GivenName -Path "CN=Users,DC=zst,DC=com"
-AccountPassword (ConvertTo-SecureString -AsPlainText $_.password
-Force) -UserPrincipalName $_.UserPrincipalName -Company $_.Company
-Department $_.Department -Enabled $true}
```

TREŚĆ ZADANIA

1. Korzystając z polecenia Idifde wyeksportować informacje o całej domenie do pliku. Potwierdzić to zrzutem ekranu z interpretera poleceń oraz z utworzonego pliku, przedstawić informacje dotyczące grupy "Administratorzy".
2. Korzystając z polecenia Idifde wyeksportować informacje o dowolnym użytkowniku do pliku, zrzuty ekranu - jak wyżej.
3. Utworzyć dwie jednostki organizacyjne: "Produkcja" i "Administracja".
4. Utworzyć w j.o. "Produkcja" za pomocą Idifde 10 włączonych kont użytkowników z takimi danymi jak: imię, nazwisko, nazwa główna konta, opis "Produkcja" – zrzuty jak wyżej.
5. Wyeksportować do pliku informacje o tej jednostce organizacyjnej.
6. Usunąć j.o. "Produkcja" za pomocą polecenia dsrm.
7. Utworzyć w j.o. "Administracja" za pomocą Idifde 10 kont użytkowników (5 włączonych i 5 wyłączonych) z takimi danymi jak: imię, nazwisko, nazwa główna konta, opis "Administracja", inicjały użytkownika, nazwa wyświetlana – zrzuty jak wyżej.
8. Wyeksportować do pliku informacje o tej jednostce organizacyjnej.
9. Usunąć j.o. "Administracja" za pomocą polecenia dsrm.
10. Utworzyć j.o. "Power".
11. Utworzyć w j.o. "Power" za pomocą PS i pliku *.csv 10 włączonych kont użytkowników z takimi danymi jak: imię, nazwisko, nazwa główna konta, opis "Power", inicjały użytkownika, nazwa wyświetlana – zrzuty jak wyżej.

1. INFORMACJE DOTYCZĄCE DOMENY

```
C:\Users\Administrator>ldifde -f info_domena.txt
Connecting to "WIN-U96UWJ4HA0-zst.com"
Logging in as current user using SSPI
Exporting directory to file info_domena.txt
Searching for entries...
Writing out entries.....
.....
251 entries exported
The command has completed successfully
C:\Users\Administrator>
```

```
info_domena - Notatnik
Plik Edycja Format Widok Pomoc
[dn: DC=zst,DC=com
changetype: add
objectClass: top
objectClass: domain
objectClass: domainDNS
distinguishedName: DC=zst,DC=com
instancetype: 5
whenCreated: 202210311321.0Z
whenChanged: 2022103124150.0Z
subRefs: DC=ForestDnsZones,DC=zst,DC=com
subRefs: DC=DomainDnsZones,DC=zst,DC=com
subRefs: CN=Configuration,DC=zst,DC=com
uSNCreated: 4099
dSAISignature:: AQAACgAAAAAAAAAAAAAAAAAAAAAA1FtyrJQqqEq31Tv3RRYpxQ--
uSNCreated: 12868
name: zst
objectGUID:: 1b00eVDAGE-w2c9s1Zr2ew--
creationTime: 133119492835879287
forceLogoff: -3221372836854775908
lockoutDuration: -18000000000
lockoutObservationWindow: -18000000000
lockoutThreshold: 0
maxPwdAge: -36288000000000
minPwdAge: -8640000000000
minPwdLength: 7
modifiedCountAtLastProm: 0
nextRid: 1000
pwdProperties: 1
pwdHistoryLength: 24
objectSid:: AQAQAAAAAAUAAAAw13BRVtzJ0ak99Rn
serverState: 1
uASCompat: 1
modifiedCount: 1
auditingPolicy: AAE=
nTlSrvDomain: 0
rIDManagerReference: CN=RID Manager$,CN=System,DC=zst,DC=com
fSMORoleOwner:
CN=NTDS Settings,CN=WIN-U96UWJ4HA0,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Configuration,DC=zst,DC=com
systemFlags: -1946157956
wellKnownObjects:
B:32:6227F0AF1FC41808E3801061588580F:CN=NTDS Quotas,DC=zst,DC=com
wellKnownObjects:
B:32:F48E92AAC777485E878E9421D53087D8:CN=Microsoft, CN=Program Data,DC=zst,DC=com
wellKnownObjects:
B:32:09460C8AE1E4AAE8F6AAEE7DA1E5A:CN=Program Data,DC=zst,DC=com
wellKnownObjects:
B:32:22B70C67D56E4EF891E9308FCA3DC1AA:CN=ForeignSecurityPrincipals,DC=zst,DC=com
wellKnownObjects:
B:32:18E2EAB8684F1102B9A08C84F79F805:CN=Deleted Objects,DC=zst,DC=com
wellKnownObjects:
```

DOTYCZĄCE TEGO ZADANIA INFORMACJE

